



Evaluating evasion strategies in zebrafish larvae

Yusheng Jiao^a , Brendan Colvert^a, Yi Man^{a,b}, Matthew J. McHenry^c , and Eva Kanso^{a,1}

Edited by Alan Hastings, University of California, Davis, CA; received November 4, 2022; accepted January 13, 2023

An effective evasion strategy allows prey to survive encounters with predators. Prey are generally thought to escape in a direction that is either random or serves to maximize the minimum distance from the predator. Here, we introduce a comprehensive approach to determine the most likely evasion strategy among multiple hypotheses and the role of biomechanical constraints on the escape response of prey fish. Through a consideration of six strategies with sensorimotor noise and previous kinematic measurements, our analysis shows that zebrafish larvae generally escape in a direction orthogonal to the predator's heading. By sensing only the predator's heading, this orthogonal strategy maximizes the distance from fast-moving predators, and, when operating within the biomechanical constraints of the escape response, it provides the best predictions of prey behavior among all alternatives. This work demonstrates a framework for resolving the strategic basis of evasion in predator–prey interactions, which could be applied to a broad diversity of animals.

predator–prey interactions | probabilistic modeling | fish C-start | fluid–structure interactions | hydrodynamics

The abilities to sense and evade predators are central to the survival of a diversity of prey species. The timing, speed, and direction of a prey's escape reflect the animal's evasion strategy, which is formulated by its neurophysiology and biomechanics (1). Despite the fundamental importance of predator encounters, resolving a prey's strategy is experimentally challenging due to the variability inherent to animal behavior. Predators vary in their approach toward prey, and the ability of the prey to respond is filtered through the environment and the animal's physiology, which may additionally introduce noise in sensing, integration, and motor response. The aims of the present study are to develop an analytical approach that is capable of resolving prey strategy from kinematic measurements and to use that approach to test classic theory on strategy in fish predator–prey interactions.

The interactions between an individual predator fish and prey fish offer a classic system for the study of evasion strategy. Fish evade predators with a stereotypical “C-start” response, characterized by the fish body bending into a preparatory C-shape, followed by a rapid acceleration as the body unfolds with largely planar motion (2). Fish escape behavior inspired an application of differential game theory to determine the optimal strategy of prey (3). The distance-optimal strategy is the solution to the “homicidal chauffeur” game where prey move in the direction that maximizes the closest distance achieved by a predator that maintains a constant velocity (4). The distance-optimal strategy has been invoked to explain the escape responses in animals as divergent as cockroaches (5), crickets (6), shrimp (7), frogs (8), salamanders (9), crabs (10), and a variety of fish species (11, 12). This strategy is generally considered the primary alternative to escaping in a random direction, known as the protean strategy, which offers the tactical benefit of confusing the predator (13–16). The present study considers whether previously measured escape kinematics in zebrafish larvae (17, 18) are consistent with distance-optimal, pure-protean, or alternative strategies.

The zebrafish (*Danio rerio*) larva is a compelling system for investigating evasion because it has served as a model for the neurophysiology and biomechanics of the C-start. Its small size, lack of pigmentation, and amenability to genetic manipulation have facilitated applications of functional imaging and optogenetics in zebrafish to observe and manipulate the sensory and motor circuits responsible for visually mediated escapes (19–21). A combination of high-speed kinematics, flow visualization, and computational fluid dynamics has revealed a comprehensive accounting of the fluid forces that propel the escape response of zebrafish larvae (22–26). We incorporate these findings into a consideration of the biomechanical constraints on the escape strategy.

We adopt a multipronged approach for testing the evasion strategy in larval zebrafish. Using a strong-inference technique (27), we mathematically define models for six strategies, both with and without sensorimotor noise (Fig. 1). We then proceed to

Significance

Prey are thought to evade predators by either moving randomly or following a distance-optimal strategy. Here, we introduce an analytical approach to evaluate multiple evasion strategies and the role of biomechanical constraints in fish evasion, based on previous kinematic measurements. Our analysis shows that larval zebrafish escape in a direction orthogonal to the perceived heading of the predator, rather than following a distance-optimal strategy. This orthogonal strategy maximizes distance from fast predators but requires less sensory processing than the classic distance-optimal strategy. Importantly, when constrained to conform to the mechanics of the fish escape response, the orthogonal strategy provides the best predictions of prey behavior. Our analytical approach has broad potential to reveal strategies in predator–prey interactions.

Author contributions: E.K. designed research; Y.J., B.C., Y.M., and E.K. performed research; Y.J., B.C., Y.M., M.J.M., and E.K. analyzed data; and Y.J., M.J.M., and E.K. wrote the paper.

The authors declare no competing interest.

This article is a PNAS Direct Submission.

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¹To whom correspondence may be addressed. Email: kanso@usc.edu.

This article contains supporting information online at <http://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2212630119/-/DCSupplemental>.

Published February 7, 2023.

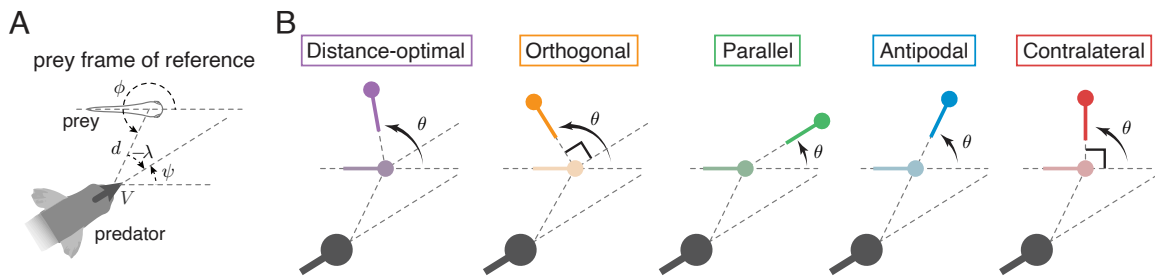


Fig. 1. Evasion strategies. (A) Schematic shows the predator position (d, ϕ) and heading ψ in the prey's frame of reference. (B) Change θ in the prey heading direction at evasion as predicted by five evasion strategies—distance-optimal: prey makes a turn that maximizes the shortest distance from the predator; orthogonal: prey turns to the direction orthogonal to the predator heading in order to flee the path of the predator; parallel: prey turns to align with the predator heading direction; antipodal: prey turns in the opposite direction of the predator angular position; and contralateral: prey turns left or right by 90° depending on the predator angular position. Strategies are distinguished by color.

evaluate these model predictions against previous measurements of escape kinematics (17) to determine the strategy that most likely describes those observations. Finally, a consideration of the escape hydrodynamics and fluid–structure interactions allows us to evaluate the constraints on these strategies. These measures combine to offer a general framework for evaluating evasion strategies in predator–prey encounters.

Results

Our description of the major results is organized around four main themes: 1) experimental data of the evasion kinematics of larval zebrafish and their descriptive statistics, 2) mathematical definitions of the evasion strategies, 3) formulation of the analytical approach and testing of evasion strategies, with and without sensorimotor noise, and 4) evaluation of the effects of biomechanical constraints on evasion.

Experimental Measurements of Escape Kinematics. We analyzed anew a large experimental dataset for the kinematics of escape responses in zebrafish larvae that were previously published (17, 18). Larvae were exposed to a robotic predator, consisting of a dead adult zebrafish (of fixed size) controlled with a motor to move through an aquarium of otherwise still water. As detailed previously (17, 18), larvae were largely motionless prior to the escape response that was stimulated by the presentation of the predator. The recorded responses of larvae were compiled from numerous experiments, each of which elicited a modest number of responses. Larvae were excluded from the analysis if they responded within a few body lengths from, or a few seconds after, another responding larva. The 3D kinematics of larvae were compiled in the predator's frame of reference to yield a cloud of responses anterior to the robotic predator.

The speed of the robotic predator was set to a constant equal to 2, 11, or 20 cm s^{-1} to reflect the speed range of a typical foraging predator (22). This ensured a repeatable stimulus that elicited a fast C-start response from the larvae (17, 18). High-speed kinematics recorded a total of 699 evasion instances: $N_{\text{slow}} = 251$ for the slow-moving predator, $N_{\text{mid}} = 233$ for the mid-speed predator, and $N_{\text{fast}} = 215$ for the fast-moving predator (Fig. 2). **Experimental analysis.** From the previous kinematic measurements, we presently calculated the predator distance d , angular position $\phi \in [0, 2\pi)$, and heading $\psi \in [-\pi, \pi)$ in the prey's frame of reference at the onset of the C-start escape response, and we calculated the change in the prey's orientation $\theta \in [-\pi, \pi)$ as it completed the C-start escape response (Fig. 1A

and *SI Appendix*, Fig. S1B). In our analysis, θ captures the rotation of the entire fish body, that is, the change in prey heading, which is not the same as the change in the body angular position employed previously (17, 18). We clearly distinguish between the prey's sensing of the predator angular position ϕ and heading ψ , which are often confused in empirical studies of evasion (12, 28, 29). In addition to the predator's actual heading direction ψ , we considered that the prey perceives λ , the deviation of the predator's heading from the angular position ϕ , given by $\lambda = \psi - (\phi + \pi)$, $\lambda \in [-\pi, \pi)$ (Fig. 1A and *SI Appendix*, Fig. S1B and C). The predatory stimulus is said to be sinistral if $\lambda > 0$; that is, the predator is headed to the left of where it appears in the prey's visual field, and dextral otherwise.

Descriptive statistics of kinematic measurements. We found no correlation between the prey's escape direction θ and its distance d from the predator at the onset of evasion (*SI Appendix*, Fig. S3). However, we did find a clear correlation between the escape direction θ and the angular position ϕ in instances where the predator appears in the prey's visual field (*SI Appendix*, Fig. S5). The data also showed a correlation between θ and the predator heading ψ , when partitioned based on whether the predator's heading is sinistral ($\lambda > 0$) or dextral ($\lambda < 0$), relative to its angular position ϕ (*SI Appendix*, Fig. S7). Importantly, although the distributions of ϕ , ψ , and θ varied with predator speed V , the correlations between θ and ϕ and between θ and ψ were qualitatively similar for all V (*SI Appendix*, Figs. S5 and S7), suggesting that for the range of speeds considered, V can be treated as a model parameter, rather than a variable that fundamentally changed the evasion behavior.

In sum, our statistical analysis (*SI Appendix*, Figs. S2–S7 and Table S1) indicates that the escape direction θ depends on the prey's sensing of the predator's angular position ϕ , heading ψ , and deviation between them λ , but does not disambiguate which stimuli determine the escape direction and the behavioral rules that best explain the data.

Definition of Evasion Strategies. We next define the six fish evasion strategies: distance-optimal, orthogonal, parallel, antipodal, contralateral, and pure-protean. We index these strategies with an integer $n = 1, \dots, 6$ in the order listed above. In all strategies, we ignore the prey biomechanics and treat both the predator and prey as point masses equipped with heading directions. Therefore, the prey's strategy is demonstrated by the direction of its escape θ . However, these escape directions vary among the strategies depending on the relative position and heading of the predator (Fig. 1B). We rate the strategies by their complexity

slow predator

mid-speed predator

fast predator

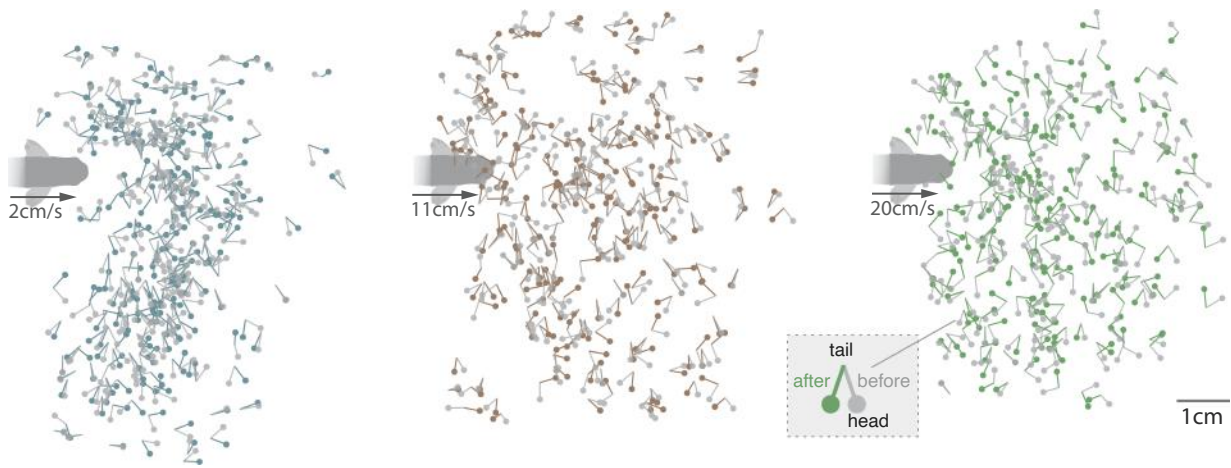


Fig. 2. Experimental measurements of zebrafish larvae evasion in response to the robotic predator, from a dorsal perspective. Zebrafish larvae were randomly placed in a tank with an approaching robotic predator driven at three speeds: $V = 2, 11, \text{ and } 20 \text{ cm s}^{-1}$. The larvae were mostly straight and motionless until exhibiting a fast C-start evasion response to the predator (17, 18). Each experiment involved a single predator–prey encounter. The experiment was repeated to collect three large datasets of size $N_{\text{slow}} = 251$, $N_{\text{mid}} = 233$, $N_{\text{fast}} = 215$ for the slow, mid-speed, and fast-moving predator, respectively. Evasion instances are superimposed for visualization purposes. For each evasion instance, we calculated, in the predator frame of reference, the position and orientation of the prey at the onset of evasion (gray macebells where the head represents the prey's position and spike represents its orientation). The change in prey's orientation θ induced by the C-start evasion response (*inset*) is shown in colored macebells. Color is used only for illustration purposes and not to be confused with the color code used in Figs. 1, 3, 4, and 6 to distinguish between evasion strategies.

of sensing (Table 1), which is a relative measure that increases with the number of geometric parameters that must be accurately determined to execute the escape in the direction predicted by the strategy. By this metric, the sensing of absolute quantities is more complex than relative quantities.

Distance-optimal evasion strategy. A distance-optimal evasion strategy considers that the prey's objective, once it detects the predator, is to maximize its minimum future distance from the predator (3, 30). Accordingly, the prey should head in the direction θ relative to its preevasion heading (SI Appendix, section 2),

$$\theta = f^{(1)}(\psi, \lambda; \chi) = \begin{cases} \psi - \chi, & \text{sinistral: } \lambda \in [0, \pi), \\ \psi + \chi, & \text{dextral: } \lambda \in (-\pi, 0), \end{cases} \quad [1]$$

where $\chi = \cos^{-1}(U/V)$ is an angle that depends on the ratio U/V of prey speed U to predator speed V . For $U > V$, $\chi = 0$. We treat χ as a model parameter rather than a variable. This distinction may not be important for the prey, but it is relevant for our subsequent analysis of this strategy.

Table 1. Sensory requirements of evasion strategies

	Predator state				Complexity of sensing
	Angular Position ϕ	Heading ψ	Heading Deviation λ	Speed V	
Distance-optimal	.	○	○	○	Most
Orthogonal	.	○	○	.	↓
Parallel	.	○	.	.	
Antipodal	○	.	.	.	Least
Contralateral	○	.	.	.	

○ Absolute value ◦ Relative value . Not needed

Orthogonal evasion strategy. We propose a simpler evasion strategy where the prey turns 90° away from the heading direction ψ of the predator,

$$\theta = f^{(2)}(\psi, \lambda) = \begin{cases} \psi - \pi/2, & \text{sinistral: } \lambda \in [0, \pi), \\ \psi + \pi/2, & \text{dextral: } \lambda \in (-\pi, 0). \end{cases} \quad [2]$$

This strategy is equivalent to the distance-optimal strategy in the fast predator limit $U/V \rightarrow 0$ but may determine θ without the need to sense the predator speed V .

Parallel evasion strategy. For a slow predator $U/V \geq 1$, the optimal strategy is for the prey to reorient itself in the direction of the predator heading, $\theta = f^{(3)}(\psi) = \psi$, which can be readily deduced by setting $\chi = 0$ in Eq. 1. The major disadvantage of this strategy is that it could place the predator in the blind spot of the prey's visual field.

Antipodal evasion strategy. Empirical observations (28, 31) suggest that the prey might follow an antipodal strategy by reorienting its heading θ in the direction opposite to the angular position ϕ , where the predator appears in its visual field, without any account for the predator heading ψ ,

$$\theta = f^{(4)}(\phi) = \begin{cases} \phi + \pi, & \text{left stimulus: } \phi \in [0, \pi), \\ \phi - \pi, & \text{right stimulus: } \phi \in (\pi, 2\pi). \end{cases} \quad [3]$$

Contralateral evasion strategy. A similar but simpler strategy, called contralateral, was suggested in (17) when the prey is approached by the predator from either side. Accordingly, the prey escapes by turning 90° either to the “left” or “right” of its own preevasion heading,

$$\theta = f^{(5)}(\phi) = \begin{cases} -\pi/2, & \text{left stimulus: } \phi \in [0, \pi), \\ \pi/2, & \text{right stimulus: } \phi \in (\pi, 2\pi). \end{cases} \quad [4]$$

Pure-protean evasion strategy. The pure-protean strategy suggests that the evasion response θ is random, independent of the predator state, with a uniform probability of moving in any particular direction. This strategy is best expressed in a probabilistic manner, where the probability distribution is uniform with constant probability density function (PDF) $p^{(6)}(\theta) = 1/(2\pi)$ of obtaining any change in orientation θ .

Testing Evasion Strategies. We developed a method for evaluating evasion strategies in terms of their ability to explain the experimental observations (Fig. 2). Although the pure-protean strategy is not supported by our experimental data (SI Appendix, Figs. S5-S7), the data exhibit some level of randomness, as indicated by the variability in the location of the predator at the onset of evasion, but potentially also due to inherent sensorimotor noise in the prey's perception of the predator and its execution of the evasion response. Our approach accounts for this variation in evaluating, comparing, and ranking the hypothetical evasion strategies. To emphasize the generality of our approach, we express it in terms of a generic stimulus $\mathbf{s}$ and response $\mathbf{r}$, without reference to the specific degrees of freedom that these vectors encompass. For the zebrafish larvae, $\mathbf{r}$ is simply θ , but $\mathbf{s}$ varies depending on the strategy; theoretically, it could encompass all or any combination of the variables that define the predator state d, ϕ, ψ, λ , and V .

To examine how well the probabilistic strategy models fit the experimental data, we interpreted the latter from a probabilistic perspective. An experimental dataset generates N samples $(\mathbf{s}_i, \mathbf{r}_i)$, $i = 1, \dots, N$, from a joint PDF, denoted by $p_o(\mathbf{s}, \mathbf{r})$, whose exact form is unknown. An evasion behavior follows a conditional PDF $p_o(\mathbf{r}|\mathbf{s}) = p_o(\mathbf{s}, \mathbf{r})/p_o(\mathbf{s})$, which is related to the joint PDF $p_o(\mathbf{s}, \mathbf{r})$ and the PDF $p_o(\mathbf{s})$ of stimuli that elicit an escape response via the law of total probability (32). Unfortunately, $p_o(\mathbf{s}, \mathbf{r})$ and $p_o(\mathbf{s})$ are unknown, and only discrete samples of these PDFs are available from experiments; thus, there is a need for further modeling and analysis.

Probabilistic models under precise vs. noisy sensing and response. We distinguish between the actual predator state $\mathbf{s}$ and the prey's

sensing $\hat{\mathbf{s}}$ of the predator state. Similarly, we distinguish between the actual escape heading $\mathbf{r}$ and the prey's desired escape heading $\hat{\mathbf{r}}$. If the prey's sensing and response are precise, we get $\hat{\mathbf{s}} = \mathbf{s}$ and $\hat{\mathbf{r}} = \mathbf{r}$. However, the sensorimotor modalities underlying evasion are often noisy: The prey may perceive a noisy version $\hat{\mathbf{s}}$ of the predator's state $\mathbf{s}$, and its desired response $\hat{\mathbf{r}}$ may be altered by noisy execution or environmental conditions to yield $\mathbf{r}$.

Each evasion strategy n , save the pure-protean, defines a desired escape response $\hat{\mathbf{r}}$ given a perceived predatory stimulus $\hat{\mathbf{s}}$ and can be expressed as a conditional PDF using the Dirac-delta function $p^{(n)}(\hat{\mathbf{r}}|\hat{\mathbf{s}}) = \delta(\hat{\mathbf{r}} - \mathbf{f}^{(n)}(\hat{\mathbf{s}}))$. The joint PDF $p^{(n)}(\mathbf{s}, \mathbf{r})$ formed based on evasion strategy n follows from the law of total probability

$$p^{(n)}(\mathbf{s}, \mathbf{r}) = \iint p(\mathbf{r}|\hat{\mathbf{r}})p^{(n)}(\hat{\mathbf{r}}|\hat{\mathbf{s}})p(\hat{\mathbf{s}}|\mathbf{s})p_o(\mathbf{s})d\hat{\mathbf{s}}d\hat{\mathbf{r}}. \quad [5]$$

Here, $p(\hat{\mathbf{s}}|\mathbf{s})$ and $p(\mathbf{r}|\hat{\mathbf{r}})$ model the noise in the prey's sensing and response. In the case of precise sensing and response, Eq. 5 reduces to

$$p^{(n)}(\mathbf{s}, \mathbf{r}) = \delta(\mathbf{r} - \mathbf{f}^{(n)}(\mathbf{s}))p_o(\mathbf{s}). \quad [6]$$

In the following, we treat each case separately.

Evaluating evasion strategies under precise sensing and response.

To obtain samples of the evasion response predicted by Eq. 6, we use as input the distribution of the empirically observed stimuli $\mathbf{s}_i$, and we construct a dataset $(\mathbf{s}_i, \mathbf{r}_i^{(n)} = \mathbf{f}^{(n)}(\mathbf{s}_i))$ for each strategy. For each predator speed, we arrive at five datasets representing theoretical predictions of the prey's evasion response according to the distance-optimal, orthogonal, parallel, antipodal, and contralateral strategies. Bivariate histograms in the (ϕ, θ) -plane for each strategy based on the dataset combining all predator speeds are shown in Fig. 3B. The histograms represent discrete cross-sections of $p^{(n)}(\mathbf{s}, \mathbf{r})$ and can be used to estimate the joint probability of obtaining a predatory stimulus ϕ_i and prey response θ_i . Compared to the histogram obtained from

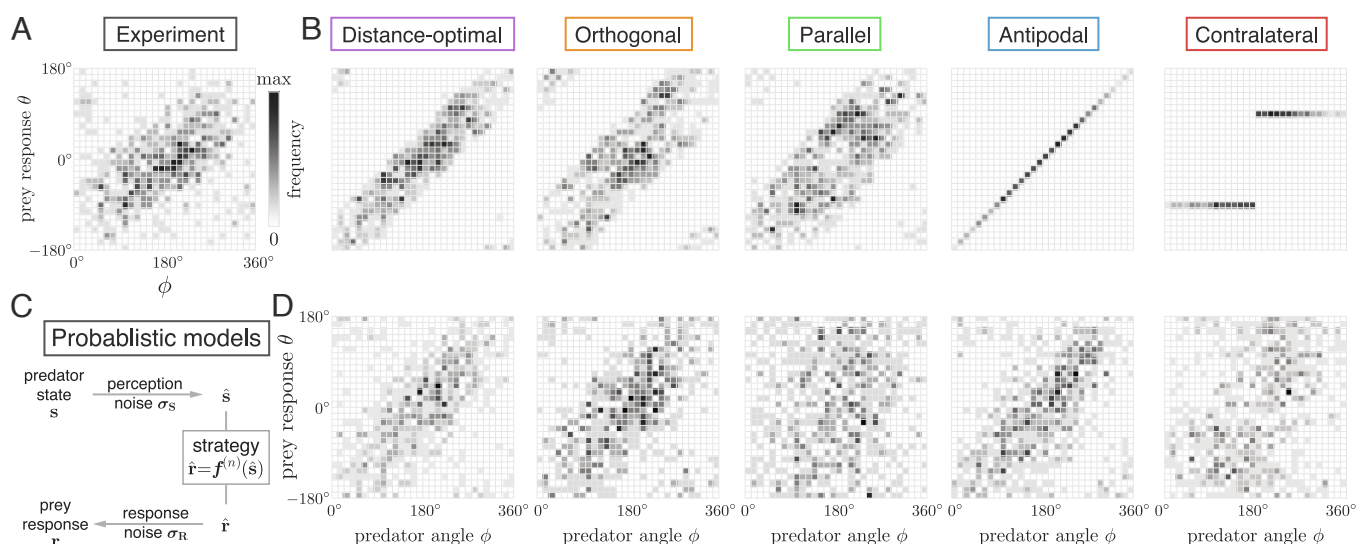


Fig. 3. Model predictions in response to experimentally observed predator states. (A) Bivariate histogram of (ϕ, θ) from experimental data. Darker color means a larger fraction of data points in that area of the (ϕ, θ) space. (B) Bivariate histogram based on the evasion models (Eqs. 1–4) with no noise; model predictions θ_i in response to experimentally observed predator states $\phi_i, \psi_i, \lambda_i, i = 1, \dots, N$, where $N = 699$ is the size of the combined data. (C) Schematic illustration of how noise in sensing and response is built into the evasion models. (D) Bivariate histogram using realizations from the noisy evasion models, Eq. 5 under optimal noise levels.